RAJKUMAR PR

Software Developer — Python

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SUMMARY

Full-stack software developer specialising in Python back-end systems, security engineering, and generative AI pipelines. Skilled in designing production-grade services for data processing, reasoning, and AI-driven applications, with strong grounding in SQL schema design, RESTful API architecture, and Git-based collaborative workflows. Proven track record delivering secure, scalable solutions across distributed, cross-geography teams, with a demonstrated appetite to grow into Node.js, API Gateway, and cloud-adjacent technologies.

TECHNICAL SKILLS

Back-End & APIs: Python (Production-Grade), Django, Flask, Django REST Framework, FastAPI, Node.js (learning), Server-Side Development

Front-End: React, TypeScript, JavaScript, HTML5, CSS3, Angular, Bootstrap, Responsive Web Design

AI & Data Processing: LangChain, LangGraph, Groq API, Llama 3, Hugging Face Embeddings, ChromaDB, Streamlit, Ollama

Databases: MySQL, PostgreSQL, MongoDB, SQLite3, Schema Design, Query Optimisation

Security & Performance: Vulnerability Assessment, Penetration Testing, Secure Code Review, Performance Optimisation

Tools & Cloud: Git, GitHub, Docker, Redis, Celery, Postman, Linux, VS Code, Cursor, API Gateway (familiar), CloudFront (familiar)

SOFTWARE ENGINEERING EXPERIENCE

Python Django Developer (Intern) — Zecser Business LLP | July 2026 – Present

- Build and maintain Django-powered web applications, contributing REST API endpoints, database models, and reusable React components across the stack.
- Work within a Git/GitHub branching workflow alongside senior engineers, pairing AI-assisted coding tools with manual code review to keep releases clean, tested, and production-ready.

Junior Software Engineer — ActiveBytes Inc. | Technopark Trivandrum | March 2025 – June 2026

- Ran end-to-end vulnerability assessments and penetration tests on client web applications and network infrastructure, translating findings into remediation reports for technical and non-technical stakeholders.
- Conducted secure code reviews across multiple client engagements, embedding AI-assisted analysis tools into day-to-day security workflows.

Python Full Stack Developer — Quest Innovative Solutions Pvt Ltd | Ernakulam, Kerala, India | July 2024 – January 2025

- Architected client-facing web applications from requirements through deployment, covering back-end logic, database integration, and HTML5/CSS3/JavaScript front ends.
- Optimised MySQL and SQLite3 schema design for faster data retrieval, resolved production security vulnerabilities, and maintained branching/merge workflows in Git.

Penetration Tester Trainee — Red Team Hacker Academy | Thiruvananthapuram, Kerala, India | October 2022 – March 2023

- Completed structured web and network security engagements spanning reconnaissance, vulnerability scanning, exploitation, and remediation reporting.
- Identified high- and critical-severity issues, including SQL injection and XSS, during simulated red-team exercises; wrote Python scripts to automate scanning tasks.

KEY ENGINEERING PROJECTS

Security Operations Center (SOC) Platform — Python, FastAPI, Uvicorn, MongoDB, PyMongo, Ollama (Llama 3.2), Django, Chart.js

- Prototyped a real-time SOC that automates threat detection, logging, and triage via an asynchronous, high-throughput ingestion pipeline feeding a local, private LLM for zero-egress analysis.
- Implemented a FastAPI/Uvicorn gateway, a regex-based signature detection engine, and MongoDB storage, plus a Django admin portal with Chart.js dashboards for live telemetry.

Agentic RAG Q&A Assistant — Python, LangChain, Pydantic, Groq (Llama 3.3), Hugging Face Embeddings, ChromaDB, Streamlit

- Built a multi-step agentic RAG assistant that retrieves, grades, and synthesises context from uploaded PDFs to produce grounded answers rather than raw retrieval output.
- Orchestrated the pipeline with LangChain and Pydantic, using Hugging Face sentence-transformer embeddings and a Groq-hosted LLM for low-hallucination, high-throughput responses.

Real-Time Collaborative Workspace — React, TypeScript, Tailwind CSS, Yjs, Django, Django Channels, DRF, Celery, Redis

- Built a Google Docs-style collaborative editor in React/TypeScript using Tiptap/ProseMirror and Yjs (CRDT) for conflict-free sync over WebSockets.
- Shipped JWT-authenticated APIs with DRF, background export jobs via Celery/Redis, and live presence/room invites through Django Channels.

Open Source SCM, Execution & File Sharing Platform — Next.js, React, Tailwind CSS, Node.js, Express.js, PostgreSQL, Prisma, Docker

- Designed an open-source code hosting platform with hierarchical project/folder management, privacy controls, and Docker-isolated remote execution with commit-style history.
- Built an encrypted secrets manager and a regex-based scanning pipeline that blocks uploads containing exposed API keys or tokens before they reach the codebase.

ADDITIONAL OPERATIONS EXPERIENCE

Data Operations & Automation Associate — Cortech Infosystem | Thiruvananthapuram, Kerala, India | October 2023 – July 2024

- Managed schema validation and digital conversion for high-volume physical datasets, structuring verification steps that reduced input discrepancies and improved archival throughput.

IT Operations & Administrative Support — Greens Mart | Thiruvananthapuram, Kerala, India | March 2023 – October 2023

- Maintained digital inventory logs and point-of-sale records, coordinating vendor communications and database entries to keep daily retail operations running smoothly.

EDUCATION

Bachelor of Engineering, Computer Science and Engineering

Narayanaguru College of Engineering, Anna University — Tamil Nadu, India | GPA: 7.5/10 | 2017 – 2021

CERTIFICATIONS

- Certified Python Full Stack Developer
- Certificate in Penetration Testing — Red Team Hacker Academy
- Infosys Virtual Experience Program; Accenture Data Analytics and Visualization

LANGUAGES

English (Full Professional) | Tamil (Native) | Malayalam (Native)